

Session state — framework tables

System instructions

the agent's standing prompt

Permission rules

permission_rule

Knowledge

knowledge — project scope + root scope

Tasks

task · task_dependency

Conversation history

turn · part — partitioned
hot / warm / cold by recency

Cross-branch summaries

branch · summarized turns

Pending state

in-flight tool calls · reserve

All rows in one MVCC store — read under one snapshot, kept consistent by the framework.

assembleContext(session, branch, budget)

six-phase assembly under a token budget

1 System instructions

2 Permission rules

3 Knowledge (project + root scope)

4 Tasks

5 History — hot / warm / cold turns

6 Cross-branch summaries ·
pending state

Packs newest-first until the budget binds;
remainder is summarized or skipped.
opts set each phase's share of the budget.

enforceBudget(scope)

prunes oldest sessions when a scope
exceeds its storage budget —
{ sessionsPruned, bytesFreed }

budget

fills

ContextWindow (returned)

blocks[] — typed, ordered

type system · permissionRules ·
knowledge · tasks · turn ·
branchSummary · pending

content

tokenCount

sourceIds (provenance)

totalTokens / budget

turnsIncluded

turnsSummarized

turnsSkipped

knowledgeFingerprint

stable while the knowledge
set is unchanged —

reuse prompt prefix /
KV cache across turns

The prompt is data with a budget and
provenance — not a truncated string.

Worked example (§7.2, measured on the published artifact) — 5 knowledge facts, 12 turns; each cell = one turn (filled = included, hollow = skipped)

budget 200  141 tokens packed · 3 turns included · 9 skipped — budget binds, oldest history evicted

budget 1000  579 tokens packed · 12 turns included · 0 skipped — all history fits

budget 4000  579 tokens packed · 12 turns included · 0 skipped — material, not budget, is now the limit

knowledgeFingerprint identical across repeated calls on the unchanged knowledge set — the reuse signal holds.